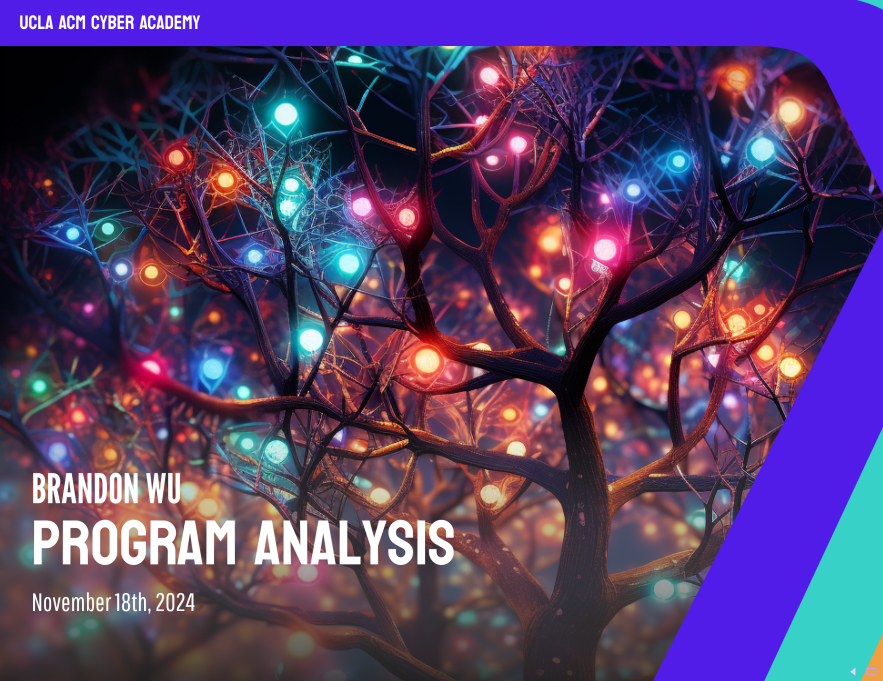


A large, dark tree with many branches is illuminated with numerous small, colorful lights in shades of red, orange, yellow, green, and blue. The lights are scattered throughout the branches, creating a festive and vibrant atmosphere. The background is dark, making the lights stand out.

BRANDON WU PROGRAM ANALYSIS

November 18th, 2024

- 1 The State of Software
- 2 Program Analysis
- 3 Implementing the Impossible
- 4 Stages of Program Analysis: Lexing
- 5 Stages of Program Analysis: Parsing
- 6 Stages of Program Analysis: IR Generation
- 7 Stages of Program Analysis: Dataflow Analysis (bonus)
- 8 Conclusions

1 - The State of Software

On August 20th, 2011, Silicon Valley venture capitalist and entrepreneur Marc Andreessen¹ published an essay entitled "Software is eating the world".

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This essay included a lot of business-oriented reasons for why software was immensely disrupting each individual economic sector, for reasons of ease of use, speed of execution, and reach of influence, among others.

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This essay included a lot of business-oriented reasons for why software was immensely disrupting each individual economic sector, for reasons of ease of use, speed of execution, and reach of influence, among others.

Now, more than a decade after this article, it's an incredibly obvious fact that software already has eaten the world. You cannot get away from it – it is everywhere, and it is everything.

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The less code we write, the less possibility of writing a bug.

So what can we say about the immense volume of code produced by the tens of millions of software developers around the world?

Answer: It is horribly, immensely buggy, and full of mistakes.

When you write a mistake in your code, what does it often look like?

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Maybe you made a typo:

X

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def fact(n):  
    if n == 0:  
        return 1  
    return n * fac(n - 1)
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```
def initialize(data):  
    d = {}  
    for s in data:  
        d[s.strip] = None  
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✗

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You have spent the past six months working on a state of the art large language model, and finally you are ready to put it to the test. You just make a few adjustments (mostly adding comments and clarifying names), before you run the model and then decide to go on a vacation to France for two weeks.

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When you return from your vacation, you discover that your model failed with:

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When you return from your vacation, you discover that your model failed with:

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This can actually happen.

What's the point? **Programmers make mistakes.**

²A programming idiom which has led to countless errors.

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There are many programmers in the world. Programs that make these kinds of silly, one-off mistakes happen millions of times in a single day. For every mistake which is caught before reaching a sensitive application, there are millions more waiting to be uncovered. We need to build tools, compilers, and programming languages that can guard against these mistakes because the cost of allowing them is too high.

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Tony Hoare once referred to his invention of the null pointer² as his *billion-dollar mistake*³. We are talking about fighting a war upon which rests not only billions upon billions of dollars across every conceivable industry, but upon which rests the security and continued operation of our society.

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Well, he tried his best.

How can we prevent these bugs? How do we make sure that, for the prodigious, gargantuan, and overflowing deluge of software that is pumped out every day, it is as safe and as correct as possible? The alternative is a reality that is horrifying to contemplate.⁴

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Software is eating the world.

It's time to bite back.

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2 - Program Analysis

My name is Brandon Wu, and I am a program analysis engineer working at [Semgrep](#).



 @onefiftyman

 LinkedIn

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Mission To profoundly improve software security.



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today, slightly differently:

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Make it **easy** to understand why doing program analysis is **hard**.

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In essence, program analysis entails **writing programs to analyze programs**.

```
def analyze(program):  
    if programIsBad(program):  
        dont()
```

Figure 1: An extremely simplified view of any program analysis tool.

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Another is that this doesn't necessarily stop the code from being written or committed in the first place. We want something even farther "left", if possible.

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- type-checking, which looks at a program and ascertains what type its constituent parts have (if any)

A computer program can be compared to a loaded pistol.

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To know whether a pistol is defective, we could just shoot it, and see whether or not it blows up. This is analogous to detecting an error at **runtime**, when actually running a given program.

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Obviously, the solution which ends up with the gun not exploding in our hands is the preferable one.

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This is just another way of saying that we need a better method than just running into errors blindly, when running code in real environments. Discovering a bug in the Curiosity rover before it takes off is a miracle. Discovering a bug in the Curiosity rover *after* it takes off is a failure.

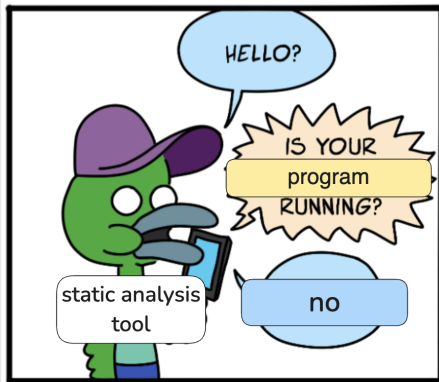
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This is the basic principle behind **shift left**, which means evaluating our programs by some criteria as far *before* runtime as possible, or farther left, chronologically.

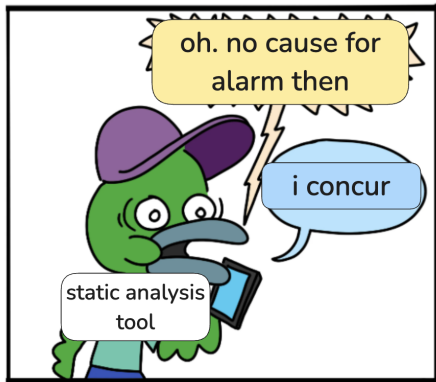
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This will motivate today's discussion of static analysis specifically, as a better way of ensuring that we find mistakes as *far left as possible*.



Bonus Panel on  PATREON



PORTUGUESE GEESE.COM

This is the mission we have ahead of us. Before we can dive into more technical details, however, we have one small issue before us:

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In English:

It is impossible to definitively answer yes or no for any property of a program's behavior, in a finite amount of time.

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This is a corollary of the [Halting Problem](#), which states that it is impossible to write a program to tell if a program loops forever or not.⁷ The reason why this is impossible come out of asking a simple question: what should the following function return?

⁷Note that these claims of impossibility are *in general*. For instance, I can look at the program `while True: pass` with my eyes and tell you that it loops forever, but we cannot write a program which does that for every program.

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This ends up producing a paradox, in much the same way as the [liar's paradox](#). Because any program can loop forever, this ends up tainting every other question that program analysis could answer, meaning that all of them are inherently impossible.

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The main innovation out of program analysis is – *it's only impossible if you insist on being right all the time.*

Generally, we want our tools to be trustworthy. The day that a pacemaker or ladder stops working is the day that someone gets hurt. Nobody wants to use something which only works *some of the time*.

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So, for most products, they must work *all the time*. To do otherwise would be sure way to head straight for bankruptcy in the free marketplace.

Program analysis suffers from no such thing. Our goal will be to implement analyses which always complete within a finite amount of time, albeit with the caveat that sometimes they might be inaccurate or incomplete, or throw their hands up and say "I don't know".

A funny corollary of this sentiment is something called the **full-employment theorem**, which essentially states that there will always be jobs in program analysis and compiler-writing, i.e. employment is always ensured⁸.

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This is because, due to the fact that the task is inherently impossible, it's always possible to write a better analysis that covers more cases, or a compiler which can produce better binaries. You just need more casework.

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You might find this demoralizing. I find it motivating, somehow.

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We have finite resources and finite time to solve a problem which is impossible to solve.

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And still, software is eating the world. The cost of failing is too high.

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And still, software is eating the world. The cost of failing is too high.

Time to put in some elbow grease and get to work.

3 - Implementing the Impossible

Consider a more specific example of program analysis, namely that of type-checking a program.

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Consider a function f which adds two ints together. We might say that it has the **type** of $(\text{int}, \text{int}) \rightarrow \text{int}$ – that is, it takes in two integers and produces an integer. Automatically determining this is the problem of type-checking.

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Def A **type** is a thing that something might be.⁹

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Consider a more specific example of program analysis, namely that of type-checking a program.

Consider a function f which adds two ints together. We might say that it has the **type** of `(int, int) -> int` – that is, it takes in two integers and produces an integer. Automatically determining this is the problem of type-checking.

Def A **type** is a thing that something might be.⁹

Let's consider the problem of giving a type to `//`, which takes two integers and computes their integer division.

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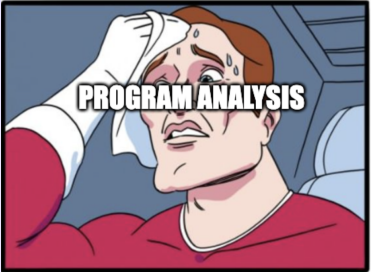
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It is impossible to have both of these things at the same time.



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So the other track will be to get the first property, but not the second. We will have to accept being wrong sometimes, and either rejecting programs which do not divide by zero, or accepting programs which do. We will call this **Type B program analysis**.

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This is the story of why everyone needs to take off their shoes in the airport.

The moral of this story is that, while it may be difficult or impossible to gather an *exact* answer (who has explosives in their shoes), it's easy to obtain an *approximative* answer: assume that *everyone* has explosives in their shoes.

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But this is precisely what a Type B program analysis purports to do. We accept that we might be wrong, sometimes, in exchange for the fact that this test always completes in short order.

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It turns out, in practice, the right solution will be to simply not care so much about the division by zero case. It's not worth the trade-offs.

We said it was impossible to have the virtues of Type A and Type B analysis at the same time. That's true, if we fix the problem statement. We might say that **Type C program analysis** is to both terminate and be correct, but at the cost of simplifying the problem we are trying to solve.

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It turns out, this is perfectly solvable in a terminating manner.¹⁰ The reason why this is OK is that "non-integer" is an approximative query – "zero" is specific.

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For most practical program analysis tools, looping forever is not an option.¹¹ So for our purposes, we are generally interested in Type B and Type C analysis.

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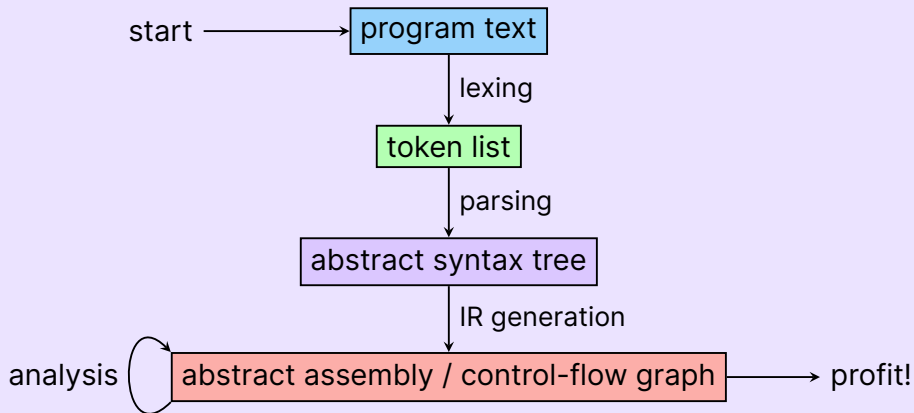
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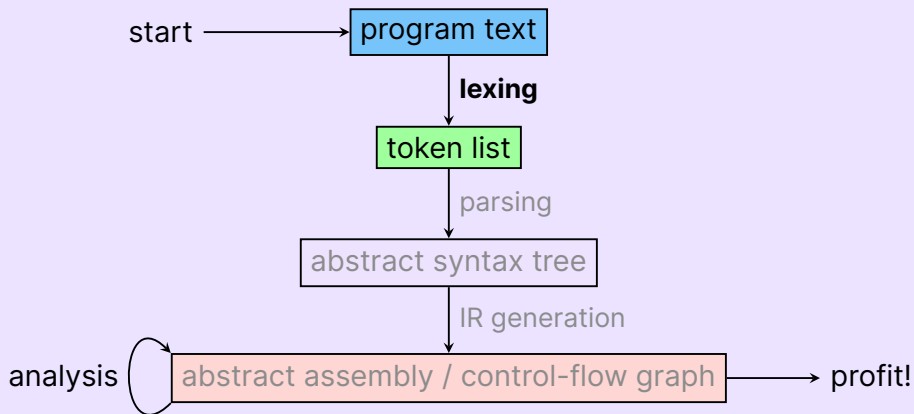
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- **analysis**, which may involve a variety of techniques, such as dataflow analysis, pointer analysis, and symbolic execution



4 - Stages of Program Analysis: Lexing



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We then **tokenize** the input program, so that instead of thinking of it as a list of characters, we group together all characters that are part of the same semantic unit.

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This is typically achieved via a **tagged union**, effectively an **enum** with data.

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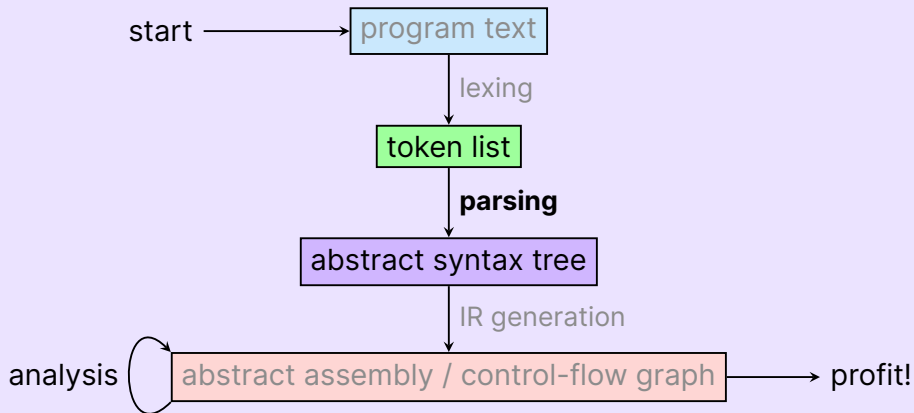
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Caveat Python is a whitespace-sensitive language, so technically this is a slightly simplified account of the information you would need to produce to parse correctly. We have elided the details here for simplicity, but the idea is the same.

5 - Stages of Program Analysis: Parsing



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Programs display a similar behavior, in that they experience a similar dual existence.
Programs can be thought of as both text and as trees.

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Cons:

- 1 difficult to analyze the semantics of
- 2 must be validated for syntactic correctness before use

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This is a simple first cut. What's wrong?

The problem is that this is only a very rough solution. It wouldn't cover, for instance, the case where we assign a variable to 0 before printing it.

```
is_zero = 0  
print(is_zero)
```

Nor would it deal with the case where we format our code slightly differently:

```
# now there's a newline before the 0
print(
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To solve this problem, we're going to need to go hug some trees.

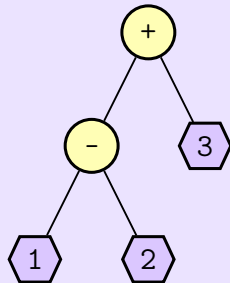
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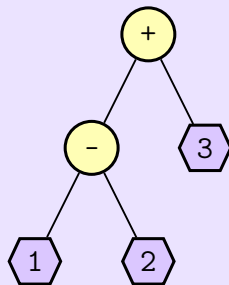
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This tree happens to denote the expression $(1 - 2) + 3$.

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In the end, this doesn't matter, because the tree structure serves as a proxy for what the parentheses were trying to tell us. We thus can get away from the precise coding details, while preserving the meaning, by using an **abstract syntax tree**, or AST for short.

We can do something very similar to op trees with programs. We will instead have an **abstract syntax tree** which denotes the structure of the program.

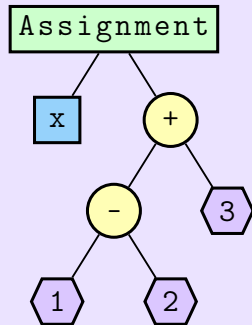
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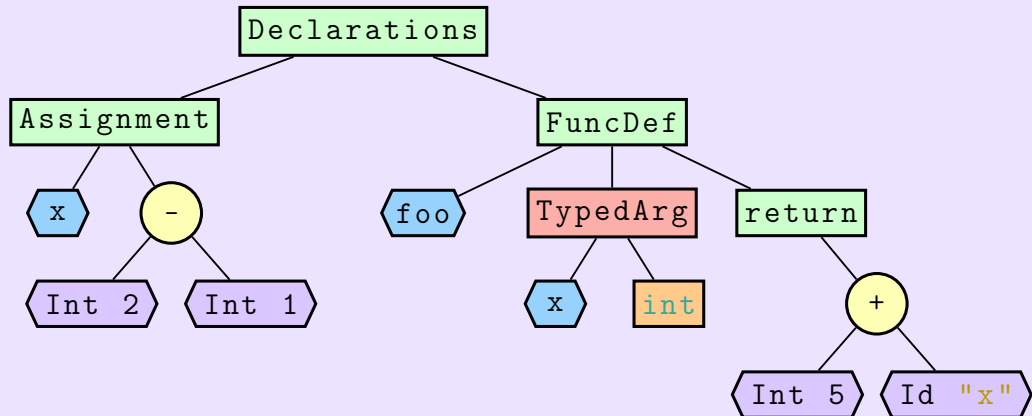
```
x = (1 - 2) + 3
```

Note how it has no mention of parens or the = sign, because they don't actually matter in terms of what the program *means*!

Generally, we can get rid of things like colons, equals signs, keywords, and parentheses in abstract syntax. These syntactic details only existed to let us know what the actual underlying tree looked like.



So, for our running example program, we could obtain the following abstract syntax tree:



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Seen in such a way, our query for finding returns of 0 can be a lot simpler:

```
def hasZeroReturn(node: ast) -> bool:
    if node.kind == "Return" and len(node.children) == 1:
        child = node.children[0]
        return child.kind == "Int" and child.value == 0
    return any(hasZeroReturn(child) for child in node.children)
```

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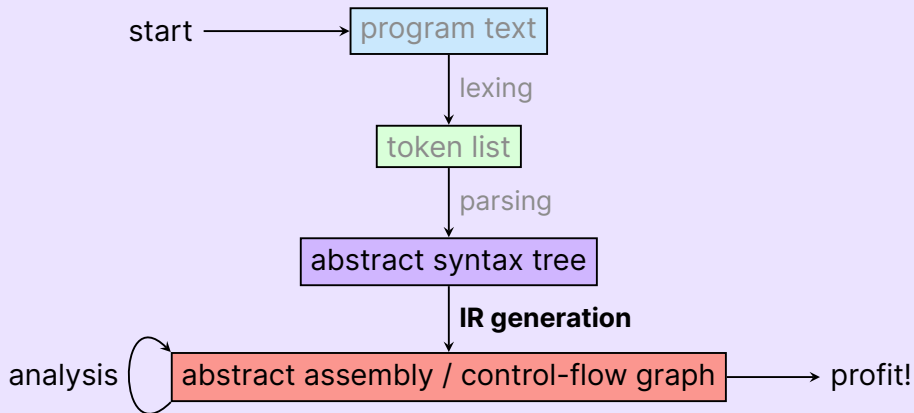
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While abstract syntax trees allow us to ignore some of the syntactic burdens of the text, they will not tell us the *entire* story of the program. We will still need to descend further, into another format, before we can answer more sophisticated questions that have to do with control-flow.

5 - Stages of Program Analysis: IR Generation



As stated before, we have some trouble with finding returns of 0 that make use of a previously-declared variable:

```
is_zero = 0  
print(is_zero)
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```

One might think that we could just iterate over statements in the program, take note of any variables which are assigned to 0, and then proceed as usual.

This would work, but we might face an issue when we see statements which are not simple variable declarations. For instance:

```
if condition:
    is_zero = 0
else:
    is_zero = 1 # haha tricked you
print(is_zero)
```

The problem is that, statically, `condition` could be a condition which is fully dynamic in nature. We might not be able to determine which branch of the `if` statement is taken, or how many times it runs.

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There are many, but a few that stand out are as follows:

- is it possible for data from some source to reach some function?
- is an error state reachable?

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    x = input()  
    y = x.strip()  
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In this code example, we see that data from the call to `input()` goes through a `strip()`, which removes whitespace, and then reaches the call to the sensitive `eval()` function.

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In this case, we find that even though `input()` does not go *directly* to `eval()`, it's still a security risk!

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```

In this case, we are interested in whether the `Error` can be reached. It's not clear, but a sophisticated program analysis tool could know that not being a Chappell Roan fan implies being a lame person, so the last condition is not actually reachable.

The commonality in all of these problems is that they are all dependent upon the **control flow** of a program.

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Because of the Halting problem, control flow is very hard to know¹⁴! In a similar vein to how we decided to solve the control-flow question of zero returns, we will restrict ourselves to two questions:

- is it **possible** for a program to exhibit a certain control-flow behavior
- is it **always** the case that a program exhibits a certain control-flow behavior

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We call these the distinction between these two questions a **may** versus **must** analysis.

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A **control-flow graph** is a kind of structure which basically describes all the paths that a program might take. When an instruction is *always* followed by another, then they appear in a single "block" together. When the instructions might *branch* (that is, go one place or another), then arrows point towards the blocks that it might lead to.

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```
x = True
y = 2

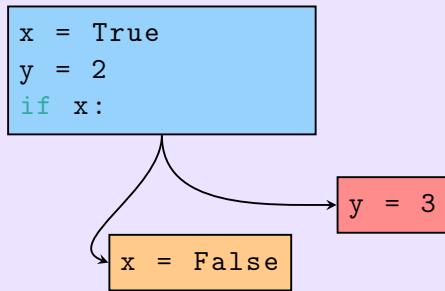
if x:
    x = False
else:
    y = 3
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Note that control-flow graphs can get pretty massive and complicated.

¹⁵Another way to think about it is that a control-flow graph describes a state space which a program may traverse over its execution. We cannot say for sure where it will go, but it must exist within the space of the graph.

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The important takeaway here is that **control-flow graphs only predict program behavior**, they do not dictate it. Because the arrows just show where a program *might* go, it doesn't magically solve the Halting Problem¹⁵.

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The important takeaway here is that **control-flow graphs only predict program behavior**, they do not dictate it. Because the arrows just show where a program *might* go, it doesn't magically solve the Halting Problem¹⁵.

The second important takeaway is that **there is always a way to turn a given program to a control-flow graph**. Let's not worry about how that happens¹⁶.

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To make sure our foundations are clear, let's look at one last example of a control-flow graph.

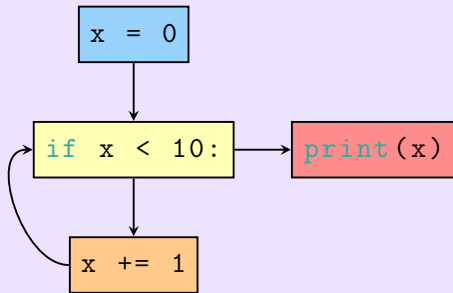
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```
x = 0
```

```
while x < 10:
```

```
    x += 1
```

```
print(x)
```

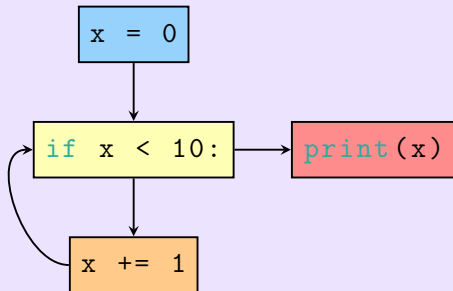


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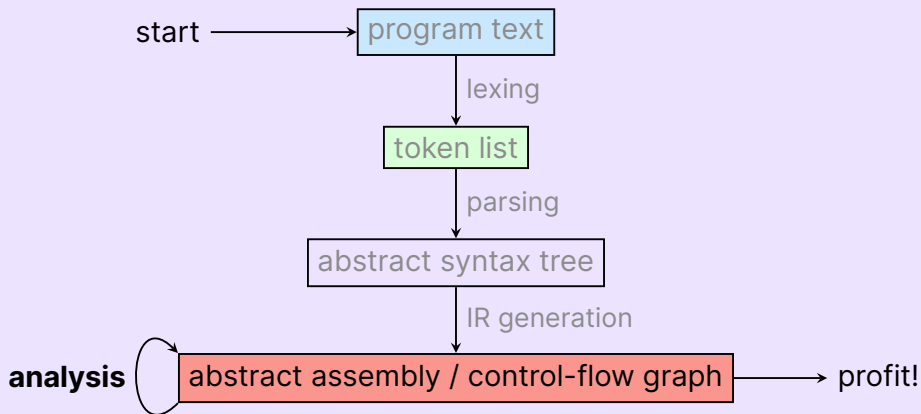
while x < 10:
    x += 1

print(x)
```



Notably, we reduce the complicated `while` construct into a an `if` statement, which leads to a node that points back to the `if` statement. This is a way of encoding the behavior of a `while` loop, which is to constantly re-calculate the condition until it is false.

6 - Stages of Program Analysis: Dataflow Analysis (bonus)



These problems all come with their own hardships. Due to what was discussed in the last section, they are all impossible to solve completely, but several of them are actually solvable via a Type B (allowed to be wrong) analysis, using the same method.

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A classic technique used in program analysis to obtain *approximative*¹⁷ answers in a finite amount of time is called **dataflow analysis**.

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A classic technique used in program analysis to obtain *approximative*¹⁷ answers in a finite amount of time is called **dataflow analysis**.

Before I can define it to you, I must give you an analogy.

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Suppose you have a query you would like to solve on programs, which has many possible answers. Further suppose that the number of possible answers is finite.

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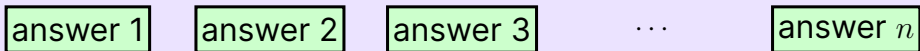
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Program analysis is hard because information can change a lot, infinitely much in fact, over the course of a program's run-time. For instance, suppose that the question is "how much memory is being used by the program?". You might pick an answer n , then move to answer $n - 2$, then move to a different answer k altogether. It's possible to jump all around, in the limit of the program's execution.

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An observation can be made that, if you can order your answer in a way such that, over the course of your analysis, you only ever change your answer by moving right, you will always eventually terminate.

This is a roundabout way of describing what is known as a **monotonic function**, which is a function which always "increases", according to some proper notion of "increases". In this case, our monotonic function also has an upper limit, i.e. a point beyond which it can no longer grow.

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For dataflow analysis, we will make use of this kind of analysis to iterate over our control-flow graph, constantly updating our answer, but only in a way that "increases", and eventually caps out. If we can do that, then we will guarantee that we will terminate.

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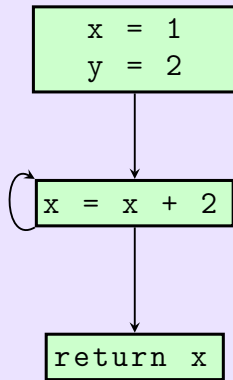
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This also usually makes our answers sometimes wrong, though.

For instance, consider the following control-flow graph¹³:

We would like to perform an analysis known as **constant propagation** on it, by noting which variables are **constant**, i.e. never changed over the duration of the program.

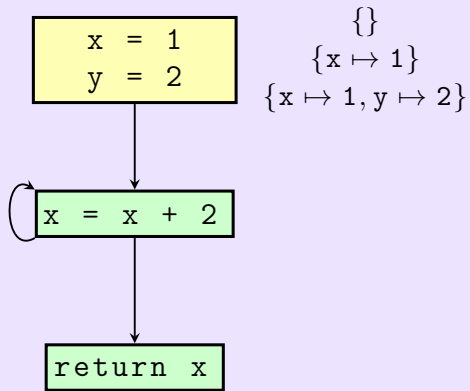
How do we do this? We simply march forward through the CFG, and noting down which variables are constant as we see them, starting with the empty set.



¹³Note that this one has a self-loop. This can happen, for example, from a `while` loop.

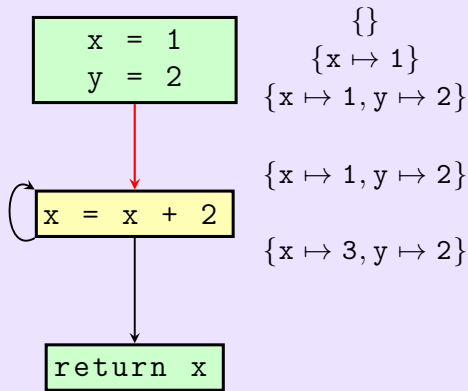
So for instance, first we traverse the entering block, by simply penciling in x and y as we see them get assigned to constants.

Once we finish, we now have the **out-set** for the first block, which we can then use to determine the other blocks.



After following the highlighted edge, we end up at the second block. Since we have some information about what variables are constant, we can carry that information here.

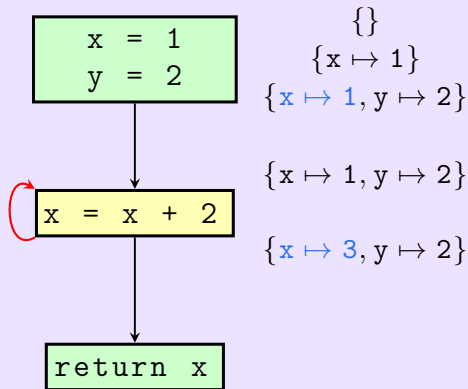
Then, we see that x is incremented by two, and thus must be constant at 3 at the conclusion of the block.



But, now we need to follow the self-loop!
Something weird happens here.

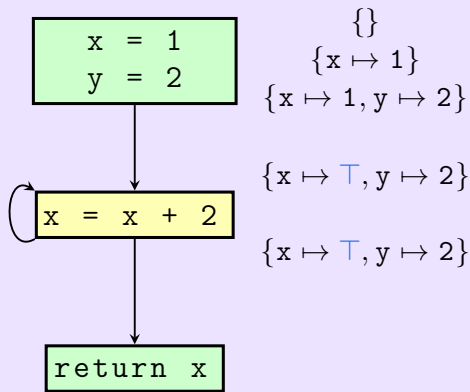
There are two conflicting out-sets that are going in to the second block. One is the one we just computed, $\{x \mapsto 3, y \mapsto 1\}$, from the output of the second block itself. The other is $\{x \mapsto 1, y \mapsto 1\}$, from the original out-set from the first block.

This means we have a conflict. x is constant, but at two different values, coming in to the second block.



This must mean that x is not constant after all.

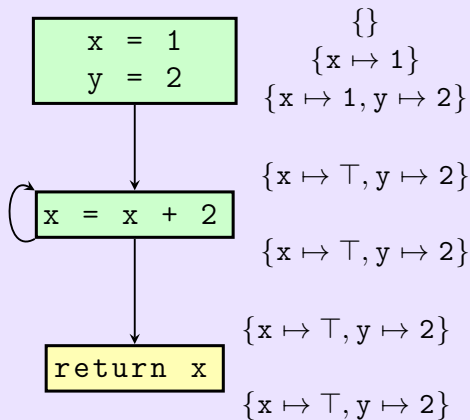
So we set the value of x to instead be $\top$, which means "not constant". Note that this is different than x not having a value, which denotes not knowing if it is constant or not.



Then, once we are assured that everything looks good, we can proceed to the final block, where we observe that we return x at a non-constant value.

This means that we cannot optimize the return value of this function after all.

Different story if we had returned y !



This is a contrived example, but the really interesting thing is that dataflow analysis works for *any* control-flow graph.

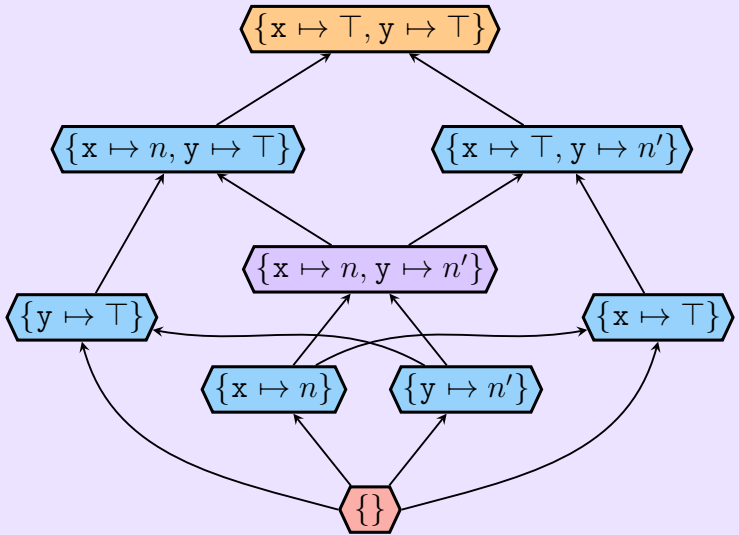
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The process seemed somewhat silly, as we could determine with our eyes that x was non-constant, but for very complicated control-flow graphs this is not an obvious fact at all. Programmatically, we can still run this same analysis, however.

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This analysis is also guaranteed to terminate, due to the monotonic reasons we stated earlier. The actual reason for this is that dataflow analysis strictly traverses up a **lattice**.



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Edges go from sets to ones which have either added a new variable at a constant, or that have upgraded a variable from a constant to the not-a-constant symbol $\top$. This represents the gaining of information, of either a variable being declared as a constant, or a variable being discovered as not-a-constant.

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- For instance, we started at $\{\}$, and then went to $\{x \mapsto 1\}$ when we read $x = 1$.
- Or, we went from $\{x \mapsto 1, y \mapsto 2\}$ to $\{x \mapsto \top, y \mapsto 2\}$ upon seeing that x was set as two different constants, along two different paths to the block.

The main thing to take away here is that *every arrow goes up*. We talked earlier about putting answers on a number line, which is represented in terms of the lattice's height, here. No matter what edge you pick, you go up some amount, which means we must terminate.

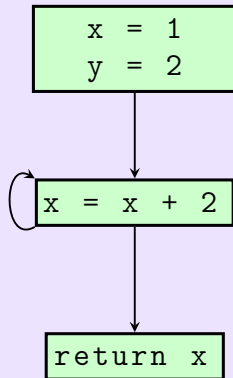
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We can't go down, because it's impossible to update your worldview to either remove a variable from the set, or to set a variable from not-constant to constant. Once you know something about a variable, you can't go back.

I mentioned earlier that this analysis is necessarily wrong, however.

The reason comes out of the fact that the control-flow graph is just how the execution of the program *might* go. In reality, it's quite possible that at runtime, we never enter the self-loop, meaning that x really is constant.

But, without running the program, we have no way of knowing, so we assume that x *is* updated at some point. This makes our knowledge necessarily possibly wrong, but a good approximation.



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The rabbit hole goes deep. This one is simple, but this is a bread-and-butter technique in program analysis.

Dataflow analysis is a foundational technique that forms the bread and butter of the program analysis toolbox. Despite being a technique which has been used for decades, it is still relevant today. All static analysis tools do some form of dataflow analysis.

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In a sense, though, it can be viewed as a "lower level" analysis. We elided a lot of details in this presentation, but a proper dataflow analysis often entails breaking down a program into its most base representation. This tends to lose a lot of the original structure of the code – nobody writes a CFG when programming.

6 - Conclusions

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Similarly, part of growing up as a security professional is learning that your tools, too, are not perfect. Throughout this lecture, we have seen that not only are they imperfect, they are fundamentally flawed.

This is not meant to inspire despair, however—quite the opposite. By understanding more the limitations of our tools, we can better understand when they are useful, and when they are not.

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Here's to making static analysis **fast**, **fruitful**, and **frictionless**.

Thank you!